

Shapes, Revisited

1.) Code:

```
def draw_ribbon(size):
    for i in range(size):
        print("'" * (i + 1) + " " * (2 * (size - i - 1)) + "'" * (i + 1))
    for i in range(size - 2, -1, -1):
        print("'" * (i + 1) + " " * (2 * (size - i - 1)) + "'" * (i + 1))

def draw_diamond(width):
    for i in range(width):
        print(" " * (width - i - 1) + "*" * (i + 1))
    for i in range(width - 2, -1, -1):
        print(" " * (width - i - 1) + "*" * (i + 1))

print("Choose a shape to draw:")
print("1. Ribbon")
print("2. Diamond")

choice = input("Enter your choice (1 or 2): ")

if choice == "1":
    size = int(input("Enter the ribbon size: "))
    draw_ribbon(size)
elif choice == "2":
    width = int(input("Enter the diamond width: "))
```

```
draw_diamond(width)
```

else:

```
print("Invalid choice!")
```

2.) Programiz Link: <https://www.programiz.com/online-compiler/1V4v1YMk5GokB>

3.) Output Example:

```
1- def draw_ribbon(size):
2-     for i in range(size):
3-         print("*" * (i + 1) + " " * (2 * (size - i - 1)) + "*" * (i + 1))
4-     for i in range(size - 2, -1, -1):
5-         print("*" * (i + 1) + " " * (2 * (size - i - 1)) + "*" * (i + 1))
6
7- def draw_diamond(width):
8-     for i in range(width):
9-         print(" " * (width - i - 1) + "*" * (i + 1))
10-    for i in range(width - 2, -1, -1):
11-        print(" " * (width - i - 1) + "*" * (i + 1))
12
13 print("Choose a shape to draw:")
14 print("1. Ribbon")
15 print("2. Diamond")
16
17 choice = input("Enter your choice (1 or 2): ")
18
19 if choice == "1":
20     size = int(input("Enter the ribbon size: "))
21     draw_ribbon(size)
22 elif choice == "2":
23     width = int(input("Enter the diamond width: "))
24     draw_diamond(width)
25 else:
26     print("Invalid choice!")
```

Choose a shape to draw:
1. Ribbon
2. Diamond
Enter your choice (1 or 2): 1
Enter the ribbon size: 3
* *
** **

** **
* *

=== Code Execution Successful ===

```

1 def draw_ribbon(size):
2     for i in range(size):
3         print("*" * (i + 1) + " " * (2 * (size - i - 1)) + "*" * (i + 1))
4     for i in range(size - 2, -1, -1):
5         print("*" * (i + 1) + " " * (2 * (size - i - 1)) + "*" * (i + 1))
6
7 def draw_diamond(width):
8     for i in range(width):
9         print(" " * (width - i - 1) + "*" * (i + 1))
10    for i in range(width - 2, -1, -1):
11        print(" " * (width - i - 1) + "*" * (i + 1))
12
13 print("Choose a shape to draw:")
14 print("1. Ribbon")
15 print("2. Diamond")
16
17 choice = input("Enter your choice (1 or 2): ")
18
19 if choice == "1":
20     size = int(input("Enter the ribbon size: "))
21     draw_ribbon(size)
22 elif choice == "2":
23     width = int(input("Enter the diamond width: "))
24     draw_diamond(width)
25 else:
26     print("Invalid choice!")

```

Choose a shape to draw:
1. Ribbon
2. Diamond
Enter your choice (1 or 2): 2
Enter the diamond width: 5

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```

=== Code Execution Successful ===